

LSH Family

* Indicates required question

1. Email *

2. Roll number *

3. Password *

Question

Suppose we have points in a 3-dimensional Euclidean space:

$p_1 = (1,2,3)$, $p_2 = (0,2,4)$, and $p_3 = (4,3,2)$. Consider the three hash functions defined by the three axes (to make our calculations very easy). Let buckets be of length a , with one bucket the interval $[0,a)$ (i.e., the set of points x such that $0 \leq x < a$), the next $[a,2a)$, the previous one $[-a,0)$, and so on.

We need to determine the candidate pairs for different bucket sizes, i.e. $a=1$ and $a=2$. For each of the three lines, assign each of the points to buckets, assuming $a=1$ and $a=2$. Determine all the correct options.

4. Options

1 point

Check all that apply.

- ☐ Only (p_1, p_2) is a candidate pair for $a=1$.
- ☐ (p_1, p_2) , (p_2, p_3) , and (p_1, p_3) are the candidate pairs for $a=2$.
- ☐ (p_1, p_2) and (p_2, p_3) are candidate pairs for both $a=1$ and $a=2$.
- ☐ There are no candidate pairs for both $a=1$ and $a=2$.